

Full Utilization of Hard-to-Recycle Mixed Plastic Waste by Conversion toward Pyrolysis Oil and BTX Aromatics on a Pilot Scale

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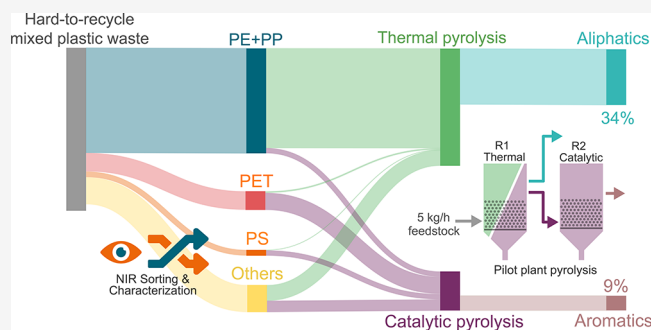


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ABSTRACT: This study investigates the technological feasibility of recycling mixed plastic waste streams into chemical building blocks on a pilot scale. Postconsumer-separated DKR-350 mixed plastic waste was first separated into two fractions by using a two-step negative near-infrared (NIR) optical sorting process. The resulting two fractions (polyolefin-rich and polyolefin-poor) were subjected to thermal and ex situ catalytic pyrolysis, respectively. Both types of pyrolysis were performed in a fluidized bed pilot plant, continuously operated at feed rates of 5 kg h⁻¹ and temperatures ranging from 460 to 550 °C. The polyolefin (PO)-rich fraction (~81 wt % PE + PP) resulted in a maximum aliphatic-rich oil yield of 48 wt % and 26 wt % gas yield based on dry material intake. The PO-poor fraction, characterized by a lower PE/PP content (~13 wt %) and high in PET and PS (43.5 and 14.4 wt %, respectively) normally unsuitable for efficient thermal pyrolysis, was processed through integrated cascading catalytic pyrolysis (ICCP) using a proprietary zeolite-based catalyst. This allowed for the successful transformation to 37 wt % aromatic-rich oil, with a total benzene, toluene, and xylenes (BTX) yield of 17 wt %, and 42 wt % gas. The high PET and PS content in the PO-poor fraction contributed to a significant increase in aromatic yields compared to the expected yield of a mixed plastic waste stream. This study thus demonstrates the potential to produce a full range of petrochemical building blocks, i.e., olefins and aromatics, from low-quality, hard-to-recycle plastic waste streams. Importantly, this work highlights that catalytic pyrolysis of the PO-poor residue stream, postsorting to yield PO-enriched streams, can effectively valorize this fraction, underscoring the technological viability of improved plastic recycling through targeted sorting and pyrolysis.



1. INTRODUCTION

The global transition toward a sustainable future is increasingly centered around the shift from fossil fuels to renewable energy sources such as solar and wind. While this transition has made significant strides in decarbonizing the energy sector, achieving true sustainability necessitates more than an energy revolution and demands a comprehensive shift in how we use and source materials. To minimize environmental impacts and foster a circular economy, materials must increasingly be derived from renewable carbon sources, such as biomass or waste, rather than virgin fossil resources.^{1,2} This paradigm shift is essential to minimize resource depletion and address the persistent challenge of reducing carbon emissions across all sectors of the economy.

Within this broader context, the challenge of plastic waste management has emerged as a critical area of focus.³ Global plastic production continues to grow, surpassing 390 million metric tons annually as of 2023, up to 359 million metric tons in 2018.^{4,5} This production growth has led to increasing

volumes of plastic waste, the majority of which ends up in landfills or the environment due to the complexity of the streams combined with the limitations of traditional recycling methods. Although incineration reduces waste volume, it comes at the cost of substantial CO₂ emissions.⁶ Developing and scaling technologies that convert waste plastics into usable materials not only diverts them from incineration but also lowers overall CO₂ emissions.⁷ However, mechanical recycling faces challenges related to degradation of plastic properties and contaminations, which result in recycled products of lower quality and limited applicability.^{8,9} Therefore, innovative recycling technologies capable of handling complex waste

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streams and producing high-quality outputs suitable for integration into existing petrochemical processes to yield recycled monomers for plastic production are desired.¹⁰

Chemical recycling, particularly pyrolysis, has emerged as a promising process for polyolefin (PO)-rich streams. Thermal pyrolysis of plastics is potentially a promising method to recycle (degraded) end-of-life and mixed plastics to an oil that can via traditional refining be used to produce monomers that are the precursors for new plastic products.^{10,11} Pyrolysis involves the thermal decomposition of organic materials in an inert atmosphere, breaking down long polymer chains into smaller, more valuable hydrocarbon products such as ethylene and propylene gases and liquids such as naphtha-range hydrocarbons and aromatics. However, direct pyrolysis of mixed plastic wastes typically yields a pyrolysis oil of low quality, primarily due to the presence of heteroatom-containing compounds.¹¹ These factors hinder the further processing of pyrolysis oil in petrochemical complexes, potentially limiting its utility and economic viability.^{11–14}

A waste stream containing mostly PO has been shown to yield large amounts of oil under the right processing conditions, and oil yields up to 95% have been reported.^{15,16} In comparison to mixed plastic waste, the pyrolysis oil of pure POs produces a higher-quality oil as there are fewer impurities and the oil content is more selective toward olefins.¹⁷ Although waste plastics consist of large amounts of PO, the presence of other polymers prevents the formation of large quantities of oil. Recent advancements in sorting and pretreatment technologies have opened new avenues for improving the quality of mixed plastic waste. For example, a two-step negative near-infrared (NIR) optical sorting process produces a PO-rich stream from mixed plastic waste, which has been shown to give improved oil yields upon thermal pyrolysis.¹⁸ The first step involves identifying and removing non-PO materials from the mixed plastic waste stream, while the second step focuses on further refining the remaining plastic waste to remove any residual non-PO plastics, resulting in a higher purity fraction.¹⁸ This process also yields a second, PO-poor waste stream, rich in non-PO materials.¹⁸ By effectively segregating the waste stream into more homogeneous fractions, the pyrolysis process can be tailored to produce higher-quality products. Moreover, the feedstock entering the pyrolysis reactor is more consistent in terms of the polymer type and contamination levels. Additionally, the PO-rich fraction typically contains fewer elemental contaminants, such as chlorine from polyvinyl chloride (PVC) or oxygen from polyethylene terephthalate (PET), reducing the formation of undesirable byproducts during pyrolysis, leaving lower demands for upgrading before use in current petroleum infrastructure.¹² Aside from studying the composition of the feedstocks for pyrolysis, determining the presence of moisture and surface contamination, i.e., dirt and organic residues, is crucial as they directly impact feedstock consistency, process efficiency, product composition, and operational stability and thus must be reduced.^{12,14}

Several studies have explored catalytic approaches to enhance the efficiency of chemical recycling by pyrolysis.^{19–21} Catalytic pyrolysis offers several advantages over thermal pyrolysis, particularly in terms of the quality of the pyrolysis oil. Catalysts enhance the breakdown of polymer chains, leading to higher yields of low-molecular-weight products, such as light hydrocarbons. Zeolite catalysts specifically promote the cracking of larger hydrocarbon molecules into smaller, more valuable ones like benzene, toluene, and xylene (BTX).^{19–21}

Producing these products directly from mixed plastic waste is particularly important due to their high demand and versatility in the chemical industry. Benzene, for example, is a precursor to a multitude of chemicals including plastics such as PS and PET. The ability to produce these aromatics from mixed plastic waste provides a sustainable source of these chemicals and enhances the economic viability of recycling processes by converting wastes into high-demand products.^{22,23}

Zeolite catalysts have been widely recognized for the production of BTX from PO and biomass-derived streams.²⁴ The stability of zeolites under harsh conditions allows them to withstand high temperatures and resist easy deactivation, making them particularly suitable for catalytic applications in which feedstock purity is often limited. In addition to their structural resilience, zeolite catalysts demonstrate tolerance to impurities, which is crucial for processing mixed waste. Studies have demonstrated that while catalytic pyrolysis with zeolite catalysts may lead to a reduction in overall product yield, it significantly enhances the composition of the output by decreasing the presence of heavy, less desirable fractions and eliminating oxygenates and N-containing compounds,^{25,26} which are often problematic in downstream petrochemical processes.

Heterogeneous fractions of mixed plastic waste, which contain a variety of polymer types and contaminants, are still able to be effectively utilized in catalytic pyrolysis processes.^{27,28} Zeolites are particularly effective in an integrated cascading catalytic pyrolysis (ICCP) process.²⁹ ICCP involves cascading pyrolysis reactions, where the pyrolysis vapors generated in one stage are further processed in a catalytic second stage to improve the quality of the desired products. Zeolite catalysts are commonly used in the catalytic stage due to their acidic properties, which facilitate crucial reactions, including dehydrogenation, cracking, and aromatization.³⁰ This is particularly advantageous for mixed plastic waste streams that under conventional pyrolysis might yield lower-quality oils with high levels of heavy fractions and contaminants.

Given the complexity of plastic waste with its varied compositions and contamination levels, achieving full recyclability remains challenging. While catalytic pyrolysis offers enhanced product composition and the ability to handle diverse feedstocks, noncatalytic pyrolysis remains important for cost-effectiveness and operational flexibility. Building on recent advancements, the present study investigates the technological feasibility of recycling complex plastic waste streams into chemical building blocks through an approach that integrates sorting techniques with (catalytic) pyrolysis, both performed at a pilot scale (5 kg h⁻¹, at a total of 25 kg per run). The overall methodology employed in this work involves a two-step NIR optical sorting process to separate the DKR-350 waste stream into two distinct fractions: a PO-rich stream, primarily composed of polyethylene (PE) and polypropylene (PP), and a residual PO-poor stream. The quality of the PO-rich stream was validated by using thermal pyrolysis in a fluidized bed reactor to produce pyrolysis oil with optional dry-washing before use. The quality of the oil from such a process should benefit from the reduction of impurities in terms of non-PE/PP plastics and other contaminants, enhancing the suitability for further processing in existing refineries. Conversely, the PO-poor residue stream was subjected to the ICCP process by using a proprietary zeolite-based catalyst to produce an oil rich in aromatics. This approach thus aims to address the challenges associated with mixed plastic waste streams,

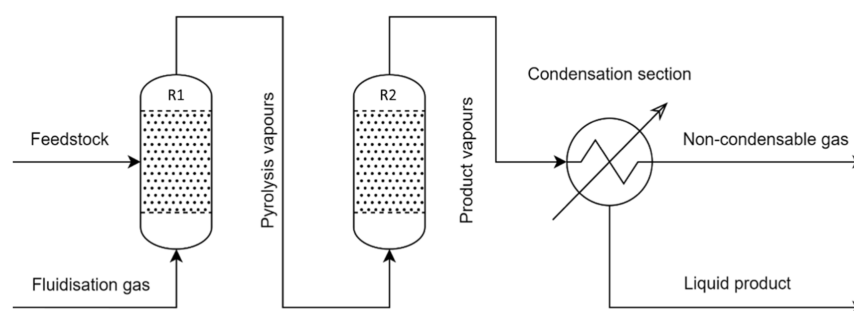


Figure 1. Schematic flow diagram of the pilot pyrolysis plant setup of BioBTX at kg-scale used in this work.

potentially offering a pathway for integrating recycling of these materials by reintegration into the petrochemical production cycle.

2. METHODOLOGY

2.1. Sorting, Characterization, and Pretreatment of Plastic Wastes. Approximately 1.4 t of postconsumer-separated DKR-350 was sourced from KSI plastic sorting facilities in Heerenveen, The Netherlands. DKR-350 is a classification³¹ used in Western Europe, particularly within the German and Dutch waste management system, to describe a specific type of mixed plastic waste predominantly collected from household packaging materials. The DKR-350 waste underwent a systematic sorting process at the NTCP sorting facility in Heerenveen (Figure S1 in the Supporting Information).³² This process was followed by shredding, dry-washing, and material composition characterization (26 categories) at NTCP, as outlined in a previous study.¹⁸ The sorting process, utilizing standard industrial equipment, included the removal of fines (<40 mm) using a vibrating screen, extraction of magnetic materials with an overhead magnet, and the enrichment of the PO stream through NIR optical sorting (Steinert Unisort PCM1000RF) to produce a fraction denoted as the “PO-rich” stream. A two-step negative optical sorting technique was employed to eliminate all materials except PE and PP. During the sorting process, non-PO materials were selectively removed, leading to the creation of a separate stream characterized as relatively low in PO, which is denoted as the “PO-poor” stream. A detailed sorting protocol is available as Section S1 in the Supporting Information.

Following the sorting procedure, the material composition of the two streams was determined by manual characterization of the output streams (two positives and one negative) using representative samples (approximately 25 kg $\times$ 3 batches). The material fractions’ weights were recorded to calculate the overall feedstock composition. From these results, the composition of the original DKR-350 material (input) was derived. All of the characterized materials were returned for further processing. Moreover, the moisture and surface contamination levels were determined (3 kg $\times$ 3 batches). Briefly, gravimetric analyses were conducted to determine the moisture content of these plastic streams (3 kg $\times$ 3 batches) using an oven at 90–95 °C for 6 h or until no change in weight was observed. The total amount of surface contamination was determined by drying the sample at 90–95 °C for a minimum of 4 h or until the sample weight stabilized. To ensure accurate measurements, the samples were distributed evenly over metal trays or appropriately divided to allow for sufficient airflow between the materials. After drying, the samples were washed

using an industrial washing machine (Laundry Lion superior cleaning model LS 135) at 85 °C with detergent and caustic soda (1%). The clean materials were dried using the same procedure and weighed. Surface contamination was quantified by calculating the difference in weight between the dirty dry material and the clean dry material.

Determining the moisture and surface contamination of plastic waste streams before and after shredding and washing is critical, particularly when preparing materials for densification and pyrolysis. To make the PO-rich stream plastic objects easier to process, they were then shredded using an industrial shredder (Figure S2 in the Supporting Information), Lindner Antares 1600 equipped with robust rotating blades and a 50 mm screen, producing the first pyrolysis feedstock, referred to as “unwashed PO-rich”. The resulting flakes were thoroughly mixed and weighed. A portion of this batch underwent dry-washing using a loop dryer (Lindner), equipped with a screen (7 mm sieve), to reduce moisture and surface contamination, yielding the second pyrolysis feedstock, termed “dry-washed PO-rich”. In this work, loop drying involved subjecting the flakes to centrifugal force (40 Hz, tip speed of 53 m s⁻¹). The constant movement of the flakes within the loop dryer helps to dislodge water and external contaminants from the surface.

After the materials were shredded into flakes, the resulting material was thoroughly homogenized through extensive mixing. The PO-poor stream was further hand-sorted for metal parts. Representative flake samples were also measured for moisture and surface contamination using the procedure outlined above. Given the unfavorable mass-to-volume ratio of the flake feedstock for pyrolysis reactor applications, the material was subjected to a densification process to enhance its suitability. This densification was conducted at the TRH Recycling facilities (Emmen, The Netherlands), using conventional agglomeration equipment. This process used frictional heat generated by mechanical action to soften the plastic, causing it to stick together. The compacted plastic flakes thus formed larger, more manageable granules, thereby increasing the bulk density of the plastic waste to make it easier to handle, store, and transport. Notably, the agglomeration temperature removed moisture but avoided inducing significant chemical alterations in the materials, preserving their original composition for subsequent pyrolytic conversion.

2.2. Pyrolysis. All noncatalytic and catalytic pyrolysis trials were conducted using a pilot-scale fluidized bed reactor at BioBTX B.V. (Groningen, The Netherlands) (Figure S3 in the Supporting Information).³³ The configuration of the pilot plant is illustrated schematically in Figure 1, whereas Table 1 summarizes the experimental conditions for noncatalytic pyrolysis used in this work. The patent of the BioBTX process describes the catalytic operation.²⁹ The pilot plant demon-

Table 1. Relevant Experimental Conditions Used in the Operation of the Pilot-Scale Pyrolysis Plant

parameter	value or range	unit
feed rate	5	kg h ⁻¹
thermal pyrolysis temperature	460–500	°C
catalytic pyrolysis temperature	550	°C

strates the integration of multiple unit operations to convert mixed plastic waste to pyrolysis oil and BTX aromatics. The feedstock, composed of agglomerated DKR-350 shredded materials, was introduced into the pyrolysis reactor via two screw feeders, ensuring a consistent material input. The first screw feeder was used as a dosing screw feeder, whereas the second one was used as a transport screw feeder. The dosing screw feeder was calibrated to ensure a continuous and known feed rate toward the transport screw. The transport screw feeder ensures an as-fast-as-possible transport of the feedstock toward the hot fluidized bed pyrolysis reactor. To prevent premature melting and sticking of the feedstock on the screw, the transport screw feeder axle and housing were actively cooled with a silicone oil. Upon introduction into the fluidized bed reactor, the feedstock was subjected to an immediate thermal treatment. Silver sand was employed as the heat transfer and pyrolysis medium as an inert material. The fluidized bed pyrolysis reactor was heated externally by using electricity. The fluidization gas flow (nitrogen) was 2.5 kg h⁻¹. A controlled feedstock throughput of about 5 kg h⁻¹ was maintained across all experiments, providing a standardized reference for the feedstock under investigation. Each experimental run utilized approximately 25 kg of feedstock. Temperature monitoring was achieved through multiple thermocouples strategically placed within the reactor. The process design included a gravity-driven discharge of solid inert materials, entrained with the pyrolysis vapors, into a catch vessel. Following this, the effluent stream was passed through a cyclone, also maintained at approximately 500 to 550 °C, which efficiently removed finer particulate matter, and subsequently collected in the catch vessel. The remaining gases and particulates were filtered through a high-temperature ceramic filter, yielding a purified gas stream.

The thermal pyrolysis experiments were conducted in a noncatalytic mode in the fluidized sand reactor “R1”, with the catalytic reactor “R2” left empty. Throughout the experiments, the continuous monitoring of gas flow and the composition of primary gas species was performed in an Agilent 990 Micro GC System. The pyrolysis vapors were initially cooled in a condenser with the resulting liquids collected in a condensation vessel. Gases escaping from the condenser were cooled further in an additional cooling step with a secondary (water-cooled) and tertiary (cooled to –10 to –20 °C) heat exchanger and then subjected to scrubbing. Clean biodiesel (Sunoil Biodiesel) was used as the scrubbing liquid, resulting in the collection of two distinct liquid streams or condensates, namely, a relatively heavy oil fraction from the condenser and a lighter fraction dissolved in biodiesel. The condensates were collected hourly over a 5 h period, which later were combined into a single container and subjected to further analysis. The solid pyrolysis residues (char) were collected from the catch vessels and weighed after each experiment. Coke yield was quantified by measuring the release of CO and CO₂ gases during the combustion of cokes in an air atmosphere following the pyrolysis experiments.

In subsequent ex situ catalytic pyrolysis experiments, the filtered pyrolysis vapors were directed to the catalytic reactor (R2), where they were converted to light olefins and aromatics. The reactor R2 contained a proprietary zeolite-based catalyst. Similar to noncatalytic runs, the hot product vapors were subsequently cooled to condense heavier fractions to separate them from the gas phase. The remaining gas passed through a series of scrubbers to remove any remaining condensable hydrocarbons and impurities. The final products were also separated and collected. Other products such as char, scrubber oil with light oil fractions, and condensate were also collected. Total coke yield was determined by measuring the release of CO and CO₂ during the combustion of the coke deposits in an air atmosphere. A separate sample was taken from the catalyst before combusting the coke deposits to regenerate the catalyst and establish the coke fraction on the catalyst. For compositional analysis, the ash content of each feedstock was determined as a measure of inert residue. Ash content was determined by combusting 80 g of each agglomerate feed material at 500 °C overnight. The resulting ash content was assumed to represent the total inorganic fraction within the complete feedstock.

2.3. Product Characterization. The yields and mass balances of the thermal and catalytic pyrolysis of the plastic waste streams were determined by agglomerate intake. The pyrolysis gas composition was analyzed using a micro-GC (Agilent 990 Micro GC System), equipped with a CP-Molsieve 5A column (20 m × 0.2 mm × 30 μm), a PoraPLOT Q column (10 m × 0.25 mm × 8 μm), a CP-Wax 52 CB column (10 m × 0.25 mm × 1.2 μm), and a CP-Sil 5 CB column (8 m × 0.15 mm × 2 μm). Following the condensation phase, additional gas products were analyzed on the micro-GC as well to gain further insights into the gas conversion processes. The gas was quantified using an internal standard (Ar) with a known flow. All quantified components were individually calibrated using a calibration gas mixture.

Liquid products were analyzed using an Agilent 8860 gas chromatograph coupled to a flame ionization detector (GC-FID). The oil composition was further characterized with an Agilent 8890 gas chromatograph coupled with a flame ionization detector and a mass spectrometer (GC-FID-MS). The carbon distribution of the oils was obtained by a modified simulated distillation method. An HP6890 GC with two columns in series (Restek MXT Siltek Guard column of dimensions 10 m × 0.28 mm and a Restek MXT-1HT film of dimensions 10 m × 0.53 mm × 2.65 μm) was cooled with liquid nitrogen to –20 °C for improved separation of volatile compounds. The retention times were calibrated against a C₇–C₄₀ calibration standard and a C₅–C₇ calibration standard. The distribution was denoted as the area percentage in each carbon fraction.

The oil was also characterized using a two-dimensional gas chromatograph with a time-of-flight mass spectrometer (GC × GC-TOFMS) for the detection of low-level impurities using an Agilent 7890B GC with JEOL AccuTOF GC 4G detector MS. The columns in the system were a Restek Rtx 1701 (30 m × 0.25 mm × 0.25 μm) primary column and a Restek Rxi-5Sil MS (1.2 m × 0.10 mm × 0.10 μm) secondary column. The secondary column was modulated at 6 s with a Da Vinci DVLS GC2 modulator. Before analysis, pyrolysis oil samples were dissolved in diethyl ether to a 2% solution. The GC × GC-TOFMS acquisition was performed after a 3.5 min solvent

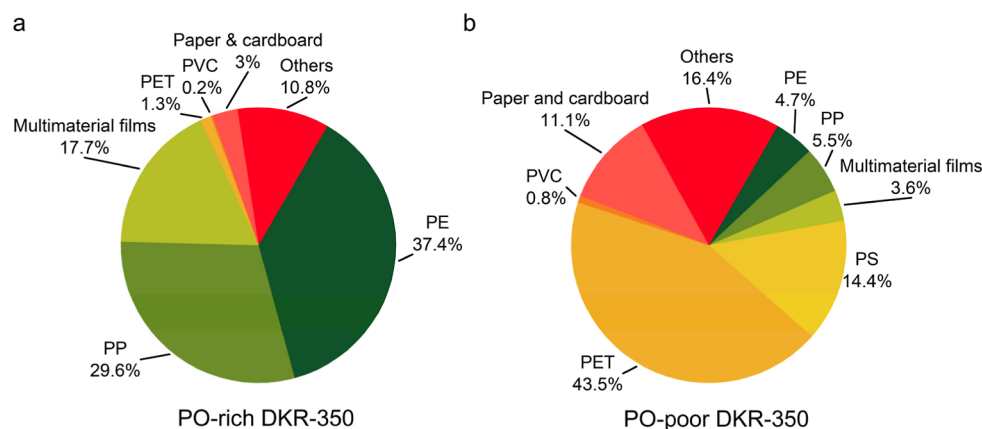


Figure 2. Material composition of (a) PO-rich and (b) PO-poor streams obtained after two-step NIR optical sorting of DKR-350 mixed plastic waste.

delay, and results were subsequently analyzed with a NIST library.

A qualitative comparison between catalytic and thermal pyrolysis oils was performed using NMR. Sample preparation involved the dissolution of the oil sample in CDCl_3 . Carbon nuclear magnetic resonance (^{13}C NMR) spectra were recorded on an Agilent spectrometer (400 MHz) using a 90° pulse and an inverse-gated decoupling sequence with a relaxation delay of 10 s, a sweep width of 225 ppm, and 1024 scans. Samples were prepared by dissolving about 100 mg of oil in CDCl_3 . Heteronuclear single quantum correlation (2D HSQC) NMR spectra were also acquired on the Agilent spectrometer with the following parameters: 11 ppm sweep width in the F2 domain (^1H), 220 ppm sweep width in the F1 domain (^{13}C), 8 scans, and 512 increments.

3. RESULTS AND DISCUSSION

3.1. Material Composition of the PO-Rich Stream and PO-Poor Residue Stream. The composition of the feedstock is a critical factor in determining the efficiency and product distribution of the pyrolysis process. Mixed plastic waste streams, such as the postconsumer-separated DKR-350 material used in this study, are particularly challenging due to their heterogeneous nature. Before NIR optical sorting, the DKR-350 material consisted of a wide variety of plastic types, including PE, PP, PS, PET, textile, paper and cardboard, organics, and PVC (see details in Table S1 in the Supporting Information). This complexity necessitates an effective sorting strategy to improve the homogeneity of the feedstock for achieving decent quality oil via thermal pyrolysis. The initial sorting stage involved the removal of materials less than 40 mm in size, which were predominantly surface contaminants, using a vibrating screen. This step eliminated 55 kg of material, accounting for approximately 4% of the total DKR-350 input by mass. Additionally, 2.4 kg of magnetic materials, representing 0.2% of the mass, were extracted using an overhead magnet. These preliminary sorting steps are essential for minimizing contamination in subsequent pretreatment steps. Following these initial steps, the DKR-350 material was further segregated into two primary streams by using a two-step negative NIR optical sorting process. The total quantity of the PO-rich stream and the PO-poor residue stream was 999 kg (mass fraction of 72%) and 333 kg (mass fraction of 24%), respectively. The separation efficiency of the NIR optical

sorting process is found in Table S2 in the Supporting Information.

The PO-rich stream is predominantly composed of PE (37.4 wt %) and PP (29.6 wt %) combined to 67 wt % of this stream, as illustrated in Figure 2a. The ubiquitous presence of objects composed of these two plastics aligns with findings from previous studies and is in line with their extensive use in packaging materials such as bags, containers, and various consumer goods.^{8,34} The PO-rich stream also contains a significant proportion of multilayer materials (18 wt %). Multilayer packaging is designed to provide enhanced barrier properties by combining different materials, often including around 70% PO from PE and PP layers. The presence of non-PO films, such as PV(D)C, PET, and EVOH among others, complicates the recycling process due to the challenges in separating the layers, which results in lower-quality recyclates.³⁵ The elemental composition of the PO-rich stream was determined previously and found to contain PO-rich DKR-350 containing approximately 76–80% C, 12–13% H, ~1% N, and 7–8% O.³⁶ This is an increase in hydrocarbon content compared to DKR-350 not subjected to NIR sorting or washing (73% C, 10% H).¹²

Objects made up of PET (1.3 wt %), PS (0.3 wt %), PVC (0.2 wt %), and other plastics (0.9 wt %) represent smaller fractions of the PO-rich stream. Components such as paper and cardboard (3 wt %), metals (1 wt %), textiles (0.3 wt %), organics (1 wt %), and fines (1 wt %) are also present, though in smaller proportions. Compounds at 4 wt % may include materials made from a combination of different plastics mixed with other substances, adding another layer of complexity. Nevertheless, the estimated high proportion of PO (about 82 wt %) in this stream suggests that this PO-rich stream is of significantly improved composition as compared to the parent material and more suited for thermal pyrolysis toward a valuable petrochemical input.

In contrast, the PO-poor residue stream, which constitutes 24 wt % of the sorted DKR-350 material, is dominated by PET (44 wt %) and PS (14 wt %), as shown in Figure 2b. The lower content of PE (5 wt %) and PP (6 wt %) indicates that the NIR optical sorting process was largely effective in separating these POs from other plastics. The presence of paper and cardboard (11 wt %), compounds (4 wt %), fines (6 wt %), textile (2 wt %), and other plastics (2 wt %) further highlights the heterogeneous nature of this residue. Such a heterogeneous stream that is low in PO is typically considered unsuitable for

regular thermal pyrolysis and thus of interest for testing in catalytic pyrolysis to produce aromatic compounds such as BTX (benzene, toluene, and xylenes).

Aside from composition, two key factors that influence the performance of plastic waste as a feedstock for pyrolysis are moisture content and surface contamination. Therefore, the effectiveness of shredding and dry-washing steps in reducing moisture and surface contamination was investigated (Table 2). The PO-rich stream in its object form exhibited a moisture

Table 2. Moisture and Surface Contamination of Unwashed and Dry-Washed PO-Rich Stream and Unwashed PO-Poor Residue Stream at Object and Flake Forms^a

stream	moisture (wt %)		surface contamination (wt %)	
	object-level	flake-level	object-level	flake-level
unwashed PO-rich	14.2	16.7	16.5	17.5
dry-washed PO-rich	n/a	7.3	n/a	6.8
unwashed PO-poor	25.4	24.1	18.9	17.4

^an/a means not applicable (dry-washed PO-rich obtained from loop drying flakes of unwashed PO-rich).

content of 14 wt %, which increased slightly to 17 wt % after being processed into flakes. The PO-poor stream objects exhibited a considerably higher moisture content of 25 wt %. This relatively high moisture is likely due to the composition of the PO-poor stream, which includes materials such as paper, cardboard, foams, textiles, and other hydrophilic contaminants. The flake forms of the PO-poor residue stream also showed comparable moisture contents, indicating that the shredding process did not significantly alter the moisture content.

The flakes from the PO-rich stream were also subjected to an additional dry-washing step to show the potential to reduce such impurities further. This is particularly advantageous in industrial settings where water is a concern and where minimizing the emission of the recycling process is a priority. Dry-washing has been shown to be effective in removing surface-level impurities without the need for water or chemical solvents.³⁶ The dry-washed PO-rich flakes showed significantly lower moisture and surface contaminants (both at 7 wt %). This substantial reduction highlights the effectiveness of the dry-washing process in removing moisture and surface contaminants from the plastic flakes. Both of these streams were tested for thermal pyrolysis to investigate whether this reduction, particularly surface contaminants, is beneficial for the product yield and composition.

The economic viability of the sorting process must be viewed in the context of the negative monetary value of the DKR-350 material itself. DKR-350, as a mixed plastic waste stream, is considered a low-value (or even a cost-burden material for waste management facilities), which would otherwise need to pay for its disposal through incineration or landfill.³⁷ In many cases, waste management companies actively seek cost-effective alternatives to disposal, making chemical recycling a potentially attractive solution for this material.

3.2. Thermal Pyrolysis of PO-Rich Stream. After densification to granules of 7.4 ± 2.9 mm, the PO-rich streams were pyrolyzed in the fluidized sand bed reactor at 460–500 °C. The mass balance data, illustrated in Figure 3, indicate a recovery range of 86–96 wt % across different experimental runs. Achieving such a mass balance on a kg-scale operation is

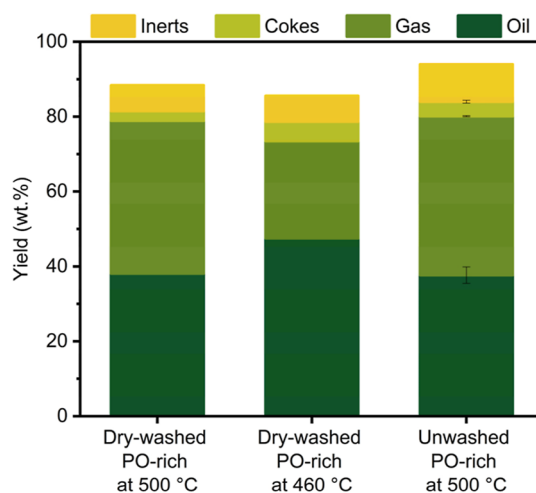


Figure 3. Product yields for the noncatalytic pyrolysis of unwashed and dry-washed PO-rich streams (see also Table S3 in the Supporting Information). Results for the unwashed PO-rich stream were taken as an average of two runs, with error bars indicating standard deviation.

within the expected range, considering the inherent complexities and practical challenges of scale-up (kg-scale) operations. Indeed, some unavoidable losses or unaccounted materials are due to factors such as minor handling losses or incomplete capture of products (e.g., some oil/wax trapped inside the pipes or reactor, some unaccounted gas products, or some minor losses during solid residue collection). The results clearly show that the pyrolysis temperature has a significant impact on the oil/wax yield. At a pyrolysis temperature of 500 °C, the dry-washed PO-rich stream produced an oil/wax yield of 38 wt %, whereas the yield increased substantially to 48 wt % when the temperature was lowered to 460 °C. This inverse relationship between temperature and oil/wax yield is well documented in the literature and can be explained by the nature of the thermal cracking process.^{18,38} At higher temperatures (i.e., 500 °C), secondary cracking reactions are more prominent. These reactions break down heavier hydrocarbons, such as wax fractions, into lighter molecules, contributing to an increase in the gas yield from 26.0 to 40.8 wt % at the expense of the oil/wax fraction. The formation of inerts (ash) was dependent on the pretreatment method with yields ranging from 7 to 10 wt % for the dry-washed and unwashed streams, respectively. The ash is predominantly composed of mineral additives, pigments, and sand, as found in previous analysis with the same feedstock.³⁶ Coke formation, although generally low, varied with the temperature and pretreatment. At 500 °C, the dry-washed stream exhibited a coke yield of 2.6 wt %, which increases to 5.2 wt % at 460 °C. However, the overall low coke content across both conditions suggests that noncatalytic pyrolysis of PO-rich streams is efficient in minimizing this undesirable byproduct, a critical factor for maintaining stable operating conditions. During thermal operation, no scaling or deposition of benzoic or terephthalic acids was observed.

The unwashed PO-rich stream gave almost the same oil/wax and gas yields as the dry-washed one (38 and 42 wt %) at 500 °C. Dry-washing of the feedstock gave a slightly lower inert fraction (7 vs 10 wt % for the unwashed) and a slightly lower coke yield (3 vs 4 wt % for the unwashed). This can be attributed to the removal of some surface contaminants during this process (Table 2). However, this reduction in inerts was

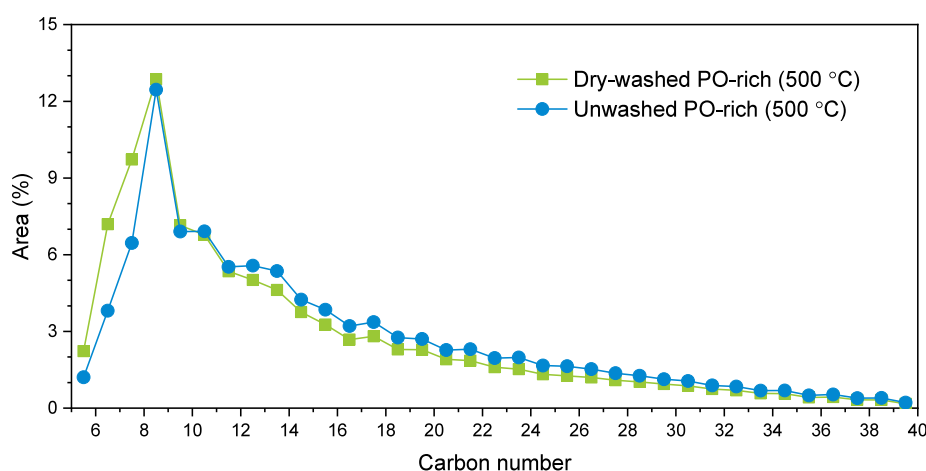


Figure 4. Carbon number distribution comparison of pyrolysis oils obtained from the thermal pyrolysis of dry-washed and unwashed PO-rich streams.

Table 3. Elemental Analysis of Nitrogen, Sulfur, Phosphorous, Silicon, and Chlorine of the Pyrolysis Oils Obtained from Thermal Pyrolysis of Dry-Washed and Unwashed PO-Rich Streams in the BioBTX Fluidized Bed Pilot Plant and Relevant Literature Comparisons

feedstock	nitrogen (wt %)	sulfur (ppmw)	phosphorus (ppmw)	silicon (ppmw)	chlorine (ppmw)
PO-rich dry-washed (460 °C)	0.7	75	9	20	155
PO-rich dry-washed (500 °C)	0.7	64	12	20	25
PO-rich unwashed (500 °C)	0.7	91	8	20	74
reference PO-rich unwashed (batch stirred tank reactor) ³⁶	0.65	300			17
reference PO-rich dry-washed (batch stirred tank reactor) ³⁶	0.99	900			32
reference DKR-350 (fixed bed reactor) ¹²	0.55	798	496	6070	1825

relatively minor, suggesting that the bulk chemical composition of the PO-rich stream (primarily PE and PP) and the process conditions dictate the overall pyrolysis results more strongly.

The results above are in line with some previous literature. For example, Kaminsky et al. used a mixed plastic stream, consisting of approximately 75% PO, to yield 48.4% oil.³⁹ However, there was a large fraction of PS present in the waste stream, and this resulted in an oil that was very rich in aromatic compounds. Del Remedio Hernandez et al. performed thermal pyrolysis on a pure HDPE stream at 500 °C in a fluidized bed reactor to obtain 48.3% waxes and oils.⁴⁰ The results presented in this work thus produce similar oil yields compared to other works using fluidized bed pyrolysis reactors.⁴¹ However, for maximizing oil yield, other reactor types have shown to be more suitable, as mixed plastic in batch reactors has shown yields exceeding 70% for waste plastics.^{28,42} The same PO-rich feedstock has also been shown to yield liquid product over 70 wt % in a batch reaction in a stirred vessel, though this was performed at a laboratory-scale only.³⁶ The latter has also shown a 10% increase in oil yield for dry-washed DKR-350 compared with unwashed DKR-350 due to a reduction in the inert material and reduced char formation. This work observes a similar trend as a slightly lower yield of char and inert is found for dry-washed PO-rich plastics (11.4 wt % for the dry-washed PO-rich feedstock compared to 13.4% and 14.2% for the unwashed PO-rich feedstock at 500 °C). Due to challenges with oil recovery in the system, the increased oil yield is not observed. As the gas yield for the unwashed feedstock is higher than the gas yield for dry-washed feedstock, there is likely an increase in oil yield for the dry-washed feedstock, but this cannot be confirmed based on these pilot runs.

The carbon number distribution was determined via simulated distillation. Figure 4 illustrates the carbon number distribution of oil obtained from pyrolysis of dry-washed and unwashed PO-rich streams at 500 °C. The distribution is presented as a fraction of the area percentage across various carbon numbers. The oil/wax from the dry-washed stream and the unwashed stream shows very similar hydrocarbon distributions, maximizing in the C₉–C₁₀ range, with the peak fraction reaching approximately 13%. There is a slightly increased light oil content (C₅–C₈) and slightly reduced heavy oil and wax content (C₁₀₊) in the dry-washed pyrolysis oil compared with the unwashed pyrolysis oil. These differences are too small to establish a significant effect of plastic pretreatment on the pyrolysis oil products. The differences are attributed to losses from heating the pyrolysis products for homogenization of the oil and wax. Further characterization of the oil was performed by 2D HSQC NMR. The dry-washed PO-rich pyrolysis oil is rich in aliphatic carbons and does not show signals typical of alcohol or ether functionalities (Figure S4 in the Supporting Information). In addition, a small amount of aromatic signals are observed in the pyrolysis oil.

Table 3 presents the elemental contaminant profiles of pyrolysis oils obtained under different treatment conditions. The data show a modest reduction in sulfur levels in the oils derived from dry-washed feedstocks compared to unwashed ones, indicating surface-level sulfur contaminants can be partially mitigated by dry-washing. However, nitrogen content remained largely unchanged across treatments, presumably because nitrogen is more fundamentally embedded within the polymer structure of the feedstock as opposed to being surface-bound. Nitrogen compounds in polymers, such as those in nylon or polyurethane, as well as certain additives, most likely

contribute to this persistence in nitrogen content despite pretreatment. This hypothesis is further supported by GC $\times$ GC-TOFMS analysis, which identified various nitrogen-containing organic compounds, including benzonitrile, caprolactam, pyrrole, and other substituted nitriles. These compounds are indicative of nitrogen-based polymers and additives as well as biomass-derived contaminants, aligning with findings from previous studies.^{43–45}

Silicon and phosphorus levels showed negligible variation between the dry-washed and unwashed samples in this work, suggesting that these elements are intrinsic to the feedstock composition rather than being external surface contamination that can be removed by dry-washing. Various metals were also analyzed, but most were detected at levels below the limit of detection (LOD) (see Table S4 in the Supporting Information), indicating low metal contamination in these pyrolysis oils. This low metal content is favorable as metal impurities can be detrimental to downstream catalytic processes, potentially causing catalyst deactivation. Thus, while dry-washing offers some reduction in sulfur levels, its limited effect on nitrogen and other elemental contaminants highlights the need for further optimization, for example, in feedstock pre- or post-treatment steps (i.e., catalytic processes).

DKR-350 that was not subjected to NIR sorting to a PO-rich stream has shown much higher elemental impurity content than the impurity contents presented in this work.¹² Unsorted DKR-350 pyrolysis oil from a fixed bed reactor yields higher S, P, Si, and Cl levels (Table 3). The dry-washed PO-rich stream on the other hand inherently contains less Cl,³⁶ which is also reflected in the direct comparison with the unwashed PO-rich pyrolysis oil from the fluidized bed reactor (74 vs 25 ppmw for unwashed and dry-washed respectively). Pyrolyzing at 500 °C compared to 460 °C in the fluidized bed reactor also reduced the Cl content of the oil. Pyrolysis of the same PO-rich feedstocks in a stirred tank reactor showed N and Cl levels comparable to the pyrolysis oil from the fluidized bed reactor but higher S levels.³⁶ The NIR sorting is therefore a promising technology to reduce the impurity content from DKR-350 for thermal pyrolysis applications.

The oil from thermal pyrolysis of PO-rich stream from DKR-350 in this work does not match refinery standards for naphtha, which typically requires a carbon range of C₅–C₁₂ and even lower N, S, Si, and Cl contents.⁴⁶ Post-treatment of the pyrolysis oil is required to achieve naphtha specifications.¹¹

Acetophenone was the most abundant heteroatom-containing contaminant in all pyrolysis oil samples, followed by benzonitrile and phenol (Table S7 in the Supporting Information). The presence of acetophenone is particularly noteworthy as it is a known degradation product of PET, while phenols likely originate from resins or biomass. The presence of these compounds suggests that even with pretreatment, certain polymer types and residual material present continue to contribute specific degradation products in the oil.

Figure 5 presents the yields of various gas components from the thermal pyrolysis of the PO-rich feedstock. Propylene is the most abundant gas component in the pyrolysis of PO-rich streams, with the highest yield of 9.8 wt %. Ethylene is also produced to some extent (maximum of 5.4 wt % for PO-rich dry-washed at 500 °C). C₄ is another significant product, with the highest yield of 8.7 wt % for the unwashed stream at 500 °C. The consistent production of ethylene, propylene, and C₄ across different conditions shows potential for further

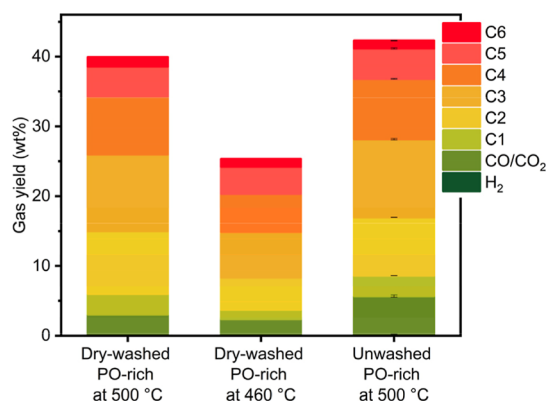


Figure 5. Gas composition from thermal pyrolysis of (dry-washed) PO-rich streams. Results for unwashed PO-rich stream were an average of two runs, with error bars indicating standard deviation.

valorization of gaseous components from the pyrolysis of the PO-rich streams. Yields of the hydrocarbon gases are reported to be lower across the whole spectrum. This result could be due to the decreased tendency of severe cracking of the products at the lower pyrolysis temperatures, resulting in, as mentioned above, higher liquid yields at the expense of the gas yield. The CO and CO₂ yields remain relatively unaffected between pyrolysis at 500 or 460 °C, originating from non-PO components in the feed. Dry-washing lowers the amount of formed CO and CO₂, likely due to the removal of surface contaminants and other oxygen-containing compounds (i.e., biogenic material).

3.3. Catalytic Pyrolysis of the PO-Poor Residue Stream. The noncatalytic pyrolysis of the PO-rich stream gave the desired results and has shown the potential of enriching waste streams in PO content to give good oil yields, highlighting the benefits of focusing on PO content in waste streams.¹⁸ Nevertheless, such enrichment provides a stream (PO-poor) that is typically unsuitable for thermal pyrolysis if high-quality hydrocarbon oils are the desired product. Consequently, the application of ICCP is advantageous as it enhances the conversion of complex waste streams into valuable aromatics and light hydrocarbons.

Figure 6 presents the mass balance data for the ex situ catalytic pyrolysis of the PO-poor stream conducted under two pyrolysis temperature conditions within the primary pyrolysis reactor (R1). Each R1 reactor condition was repeated with the same zeolite-based catalyst in the secondary catalytic reactor (R2), followed by catalyst regeneration via coke burn-off. This regeneration step, involving the controlled combustion of coke deposits, produced quantifiable amounts of CO and CO₂, allowing for the accurate calculation of coke formation. Ensuring catalyst regeneration between runs is critical for maintaining catalytic performance and preventing deactivation due to coke build-up over time.⁴⁷ The consistency in the coke yield across runs suggests that the zeolite-based catalyst remains relatively stable under the tested conditions, but routine regeneration may be necessary to sustain catalytic activity. Thus, the data from these ICCP trials affirm that while PO-poor streams may be unsuitable for thermal pyrolysis, the use of catalytic pyrolysis, particularly with a zeolite-based catalyst, offers a robust pathway to process these challenging streams efficiently.

The overall mass balance recovery for the catalytic pyrolysis of the PO-poor stream ranged from 85.6% to 96.4%, which

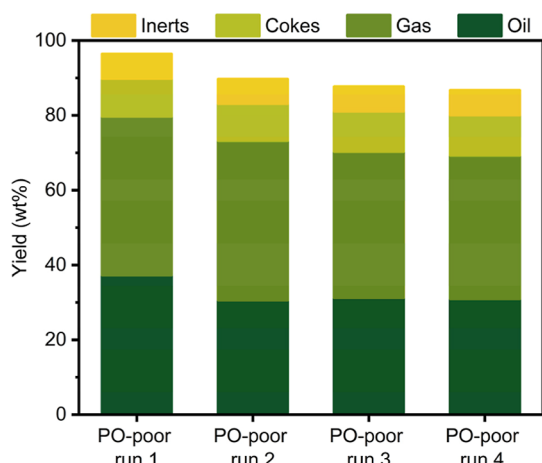


Figure 6. Product yields for the ex situ catalytic pyrolysis of unwashed PO-poor streams of subsequent runs using a regenerated zeolite-based catalyst.

falls within the expected range for pilot-scale pyrolysis, especially when considering potential losses inherent to large-scale operations. Interestingly, the data reveal no significant relationship between the pyrolysis temperature in R1 and oil yield as the oil yield slightly decreases with subsequent runs. The overall oil yield was found to be between 31% and 38% wt %. This stability indicates that the catalyst maintains its effectiveness in converting pyrolysis vapors to oil.

The coke formation was stable over each run, averaging around 10%, indicating that while coke deposits accumulate on the catalyst, their formation remains predictable and manageable. The inert content, measured at 6.6 wt %, reflects residual ash and unreacted solids, which are typical byproducts in the pyrolysis of mixed plastic waste streams.

The liquid products obtained from the ex situ catalytic pyrolysis of the PO-poor residue stream were characterized mainly for the aromatic compounds in the oil. Analysis revealed that the PO-poor stream consistently yielded high levels of aromatic compounds, predominantly monoaromatics and naphthalene derivatives (Figure 7a). This aligns with previous findings; for example, López et al. reported that catalytic pyrolysis oils derived from plastic waste were approximately 95% aromatic in composition.^{47,48} The high aromaticity in this study is further supported by 2D HSQC NMR analysis, which shows substantial aromatic functionality ($C_{ar}-H$) within the oil, predominantly from alkyl-substituted aromatic rings, with minimal oxygenated functionalities (Figure S5 in the Supporting Information).

The BTX fraction was notably high, with yields on a material intake basis ranging between 17 and 19 wt % (Figure 7b). Specifically, benzene yield was found to range from 9 to 11 wt %, toluene at 5–6 wt %, and xylene at 2 wt %. These yields compare favorably with, and in some cases exceed, BTX yields reported for catalytic pyrolysis of virgin plastics. For example, Zhang et al. previously reported BTX yields of 17% from a physical mixture of HDPE and LDPE with H-ZSM-5,

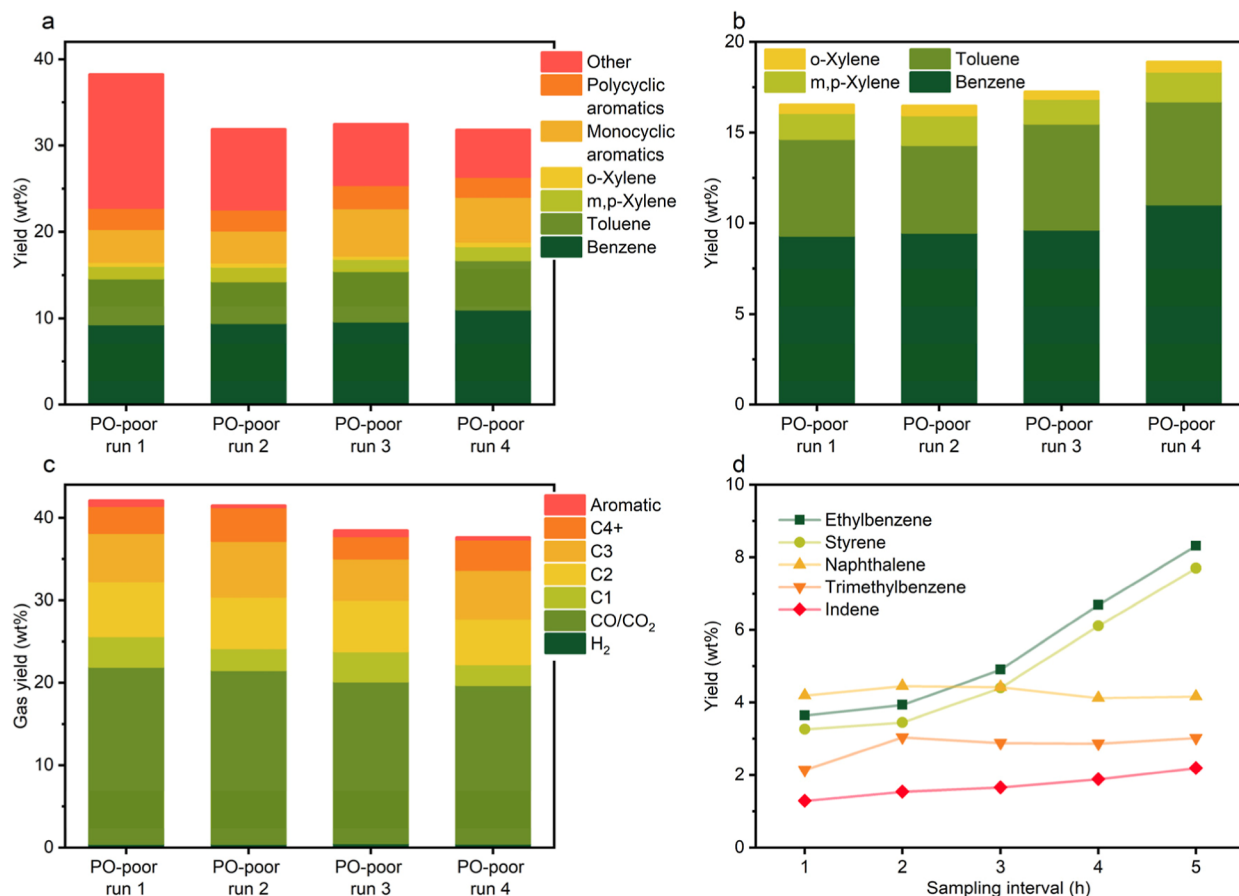


Figure 7. Yields (on material intake) of the (a) overall oil composition, (b) BTX yield, (c) product gas components for the ex situ catalytic pyrolysis of the PO-poor stream, and (d) selected other aromatic yields over time (run 1).

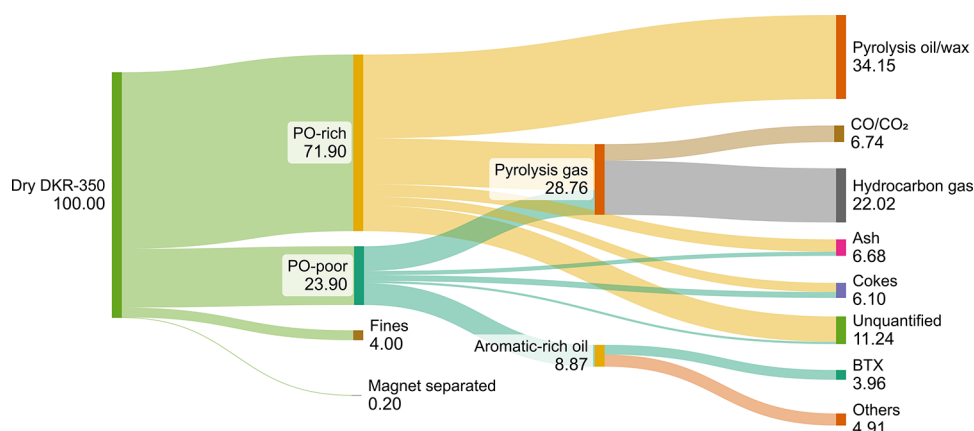


Figure 8. Material flow and distribution of feedstock components starting from DKR-350 in a pyrolysis process involving two distinct feedstock streams and their respective outputs as oil/wax, gas, aromatics, cokes, char, and other fractions (based on 100 kg of dry DKR-350).

achieving a similar selectivity of 45%.⁴⁹ Jung et al. studied the products of catalytic pyrolysis with H-ZSM-5 of PP and LDPE in a fluidized pyrolysis plant and found 7.1% BTX for PE and 5.7% BTX for PP at 668 and 660 °C, respectively.⁵⁰ Marcilla et al. examined catalytic pyrolysis of LDPE and HDPE in a microreactor and obtained an 8–9% total aromatic yield.⁵¹ The ICCP concept applied to the PO-poor stream shows to be a promising route to obtain high quantities of renewable aromatics from challenging feedstocks.

The distribution of BTX within the oil is noteworthy. While catalytic pyrolysis of PO using a zeolite like H-ZSM-5⁵² typically yields toluene as the dominant aromatic product, the higher benzene levels observed here may be attributed to the increased presence of aromatic-containing polymers such as PS and PET, in the PO-poor feedstock. This observation aligns with previous studies, for example, Yoshioka et al.⁴¹ reported that polymers with aromatic monomers tend to increase benzene production in catalytic pyrolysis.

In the noncondensable gas products, lower hydrocarbon yields were observed at reduced pyrolysis temperatures, a trend likely influenced by the minimal PO content in the feedstock. PET and PS, the primary constituents of the PO-poor stream, are less susceptible to forming light hydrocarbons compared to POs.^{53,54} Another significant finding was the high yield of oxygen-containing gases, with CO and CO₂ together accounting for 19–22 wt % of the total noncondensable gas, of which CO₂ was the dominant component. This substantial oxygen presence likely originates from PET, biogenic materials, and surface contaminations.^{55,56}

Over the course of multiple runs, a noticeable increase in ethylbenzene and styrene concentrations was observed in production of the oil products (run 1, Figure 7d). The increase in styrene content is indicative of PS degradation within the feedstock, and its accumulation suggests progressive deactivation of the zeolite catalyst. Styrene is thought to be initially converted by the catalyst into other substituted monoaromatics, such as benzene and α -methyl styrene, but as the catalyst deactivates, conversion efficiency diminishes.²⁷ For sustained industrial applications, continuous regeneration of the catalyst via periodic combustion of coke deposits would be necessary to maintain a stable product quality. Thus, this work shows the potential of the ICCP process for valorizing hard-to-recycle mixed plastic waste into valuable products.

Moreover, the presence of aromatic precursors, especially from plastics like PS and PET, enhances the yield of BTX and

other aromatic compounds.⁴¹ Thermal pyrolysis of PS typically yields high amounts of styrene, and the introduction of a zeolite catalyst partially converts styrene to other aromatic compounds. PET, on the other hand, has more diverse pyrolysis products. Initially, it is thought that PET pyrolysis yields terephthalic acid, benzoic acid, acetophenone, and other oxygen-containing derivatives, together with CO and CO₂.^{56,57} More severe pyrolysis leads to the formation of aromatic monomers and more CO and CO₂. The zeolite further aids the decomposition of oxygenated compounds to aromatics such as benzene and toluene.^{42,54}

To understand the chemical composition of the pyrolysis oil further, heteroatom-containing compounds were analyzed using GC $\times$ GC-TOFMS. Benzonitrile and benzophenone were the most abundant heteroatom-containing compounds (Table S7 in the Supporting Information). Benzophenone is a known degradation product of PET, and benzonitrile is formed after polyurethane degradation; both are also commonly used in multilayer packaging films. Additional compounds such as phenol, cresol, and dihydrosyringenin were identified and are traceable to resin-based multilayer films or biomass sources.^{57–59} Metals in the pyrolysis oil were also assessed, though most remained below the detection limit (Table S4 in the Supporting Information). Silicon was the only metallic contaminant detected at a concentration level of 7 ppmw.

3.4. Comparison between Thermal and Ex Situ Catalytic Pyrolysis Product Yields and Implications for Industrial Applications. The advent of fluidized bed pyrolysis reactors has further enhanced the efficiency and scalability of pyrolysis processes.⁶⁰ These reactors offer improved heat and mass transfer, leading to a more uniform temperature distribution and rapid reaction rates. This technology is particularly effective in ensuring consistent product yields, especially when processing mixed plastic waste streams. Moreover, the flexibility of fluidized bed reactors allows for the integration of various catalysts, optimizing the production of specific hydrocarbons depending on the feedstock composition.²⁹

Overall, the BTX yields underscore the efficiency of the ex situ catalytic process in converting a PO-depleted complex plastic sorting residue rich in aromatic structures into simpler BTX compounds. This can significantly add to the overall valorization of the complex waste stream in addition to the oil generated from the PO-rich stream. Figure 8 shows a clear visualization of how the initial mixed plastic waste is divided

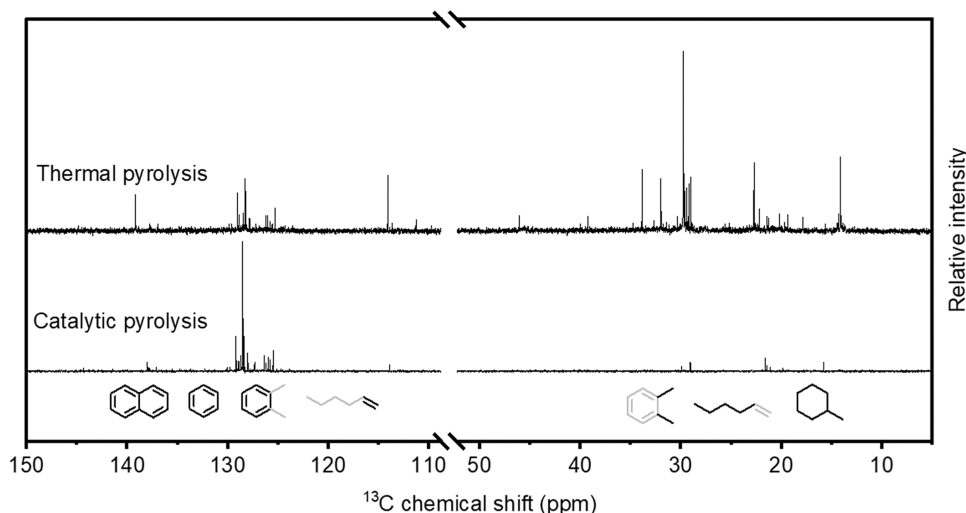


Figure 9. ^{13}C NMR of thermal pyrolysis oil (dry-washed PO-rich stream at 500 °C) and catalytic pyrolysis oil (PO-poor residue stream).

into two distinct streams. It also highlights the separation of fine and magnetically separable materials before pyrolysis, which helps eliminate contaminants and other components. When analyzing the differences between the thermal pyrolysis of PO-rich streams and the ex situ catalytic pyrolysis of PO-poor streams, several key observations and insights emerge, particularly concerning the overall yield, product distribution, and potential industrial applications. In terms of overall oil/wax yield, the ex situ catalytic pyrolysis of PO-poor streams generally produces lower oil yields compared to noncatalytic ones. However, this oil is rich in valuable aromatic compounds, including BTX. This difference highlights the influence of both feedstock composition and the presence of catalysts on the product distribution. The PO-rich streams, dominated by PE and PP, are more conducive to the production of liquid hydrocarbons, especially when lower temperatures are used in the noncatalytic process. In contrast, residual PO-poor streams contain more complex plastics, which under ex situ catalytic conditions lead to the production of more gas and coke at the expense of liquid oil.

Figure 9 shows that the ^{13}C NMR allows for a qualitative comparison between the whole oils from the thermal and catalytic pyrolysis of dry-washed PO-rich and unwashed PO-poor streams, respectively. The peaks between δ 120 and δ 150 ppm correspond to carbons in aromatic structures, such as BTX and other aromatic rings. The peaks in the catalytic pyrolysis spectrum are sharper and more intense, indicating a higher concentration of aromatic compounds. Significant signals are present between δ 10 and δ 40 ppm, indicating the presence of aliphatic hydrocarbons. The catalytic pyrolysis spectrum shows less intense signals in this region compared to the thermal pyrolysis spectrum.

The higher gas yields in ex situ catalytic pyrolysis are attributed to the role of the catalyst in enhancing the aromatization and cracking of larger molecules into aromatics and lighter gaseous products. Additionally, the composition of the PO-poor stream, which includes more aromatic-rich polymers, contributes to the generation of gases, due to their molecular composition (PET for example). The increased coke formation in ex situ catalytic pyrolysis could be due to the polymerization and carbonization of aromatic compounds, which are more prevalent in PO-poor streams (they inherently have a higher aromatic content), leading to more solid carbon

residues during pyrolysis. While coke formation is often seen as a disadvantage due to its potential to deactivate catalysts and cause operational issues such as reactor fouling, it also presents opportunities for managing and utilizing this byproduct effectively (e.g., coke as a fuel source for energy recovery or gasified to produce syngas).

These results indicate that the choice of feedstock and the use of catalysts significantly impact the yield and composition of the products. PO-rich streams are more suitable for processes aiming to maximize hydrocarbon oil production, particularly when catalysts are not used and when lower temperatures are employed. This makes thermal pyrolysis of PO-rich streams an attractive option for industries focused on producing liquid oil or chemical feedstocks that require minimal postprocessing. On the other hand, the catalytic pyrolysis of PO-poor residue streams, despite producing lower oil yields, offers the advantage of generating higher quantities of valuable gases and aromatics and offers a way to valorize a feedstock which would not be processable in conventional pyrolysis processes toward naphtha products. This makes it well suited for applications where the goal is to produce specific gaseous products or where the focus is on the production of high-value aromatic compounds. Additionally, the ability to manage and potentially utilize coke byproducts adds another layer of flexibility and economic viability to the catalytic pyrolysis process. Also, it enables one to valorize a feedstock not easily processed in chemical recycling or recycling in general.

4. CONCLUSIONS

This present study investigates the technological feasibility of recycling complex plastic waste streams (DKR-350) into valuable chemical building blocks through a combination of NIR optical sorting techniques and thermal and catalytic pyrolysis. The results demonstrate that the pyrolysis temperature is a critical factor influencing the yield and composition of pyrolysis products from PO-rich DKR-350 streams. Lowering the pyrolysis temperature, particularly to around 460 °C, significantly increased the oil/wax yield from 38% to 48% (on plastic intake), with the product being primarily wax. This outcome is advantageous for processes targeting high liquid yields for further refining. In contrast, higher pyrolysis temperatures favor gas production and produce a lighter oil

fraction, making the process more suitable for operations where lighter hydrocarbons or gaseous products are desired. Therefore, the choice of pyrolysis temperature should be aligned with the specific product distribution goals of the industry. Additionally, dry-washing the PO-rich DKR-350 stream effectively reduced surface contamination, which minimized char and inert material formation during pyrolysis, which is an operational advantage in terms of reduced maintenance and reactor fouling.

The ex situ catalytic pyrolysis of the PO-poor residue streams using a zeolite-based catalyst revealed the importance of feedstock composition and catalyst selection in optimizing the product slate. Catalytic pyrolysis of PO-poor fractions, which contain high concentrations of PS and PET, produced substantial yields of aromatics, with BTX accounting for up to 19 wt % of the material intake and approximately 60% of the total oil yield. This highlights the effectiveness of the catalyst in promoting the aromatic hydrocarbon formation. Additionally, catalytic pyrolysis also led to the formation of CO and CO₂ (approximately 20%), indicating the successful removal of oxygen-containing compounds from the feedstock, which is particularly relevant for processing oxygen-rich materials like PET.

The successful operation of the fluidized bed pilot plant validates the scalability of both thermal and catalytic pyrolysis, providing an industrially viable pathway for converting low-quality, hard-to-recycle plastic waste into valuable chemical products. These findings demonstrate that by employing sorting and selecting the appropriate pyrolysis method, even the most challenging mixed plastic waste streams can be effectively recycled. Future research should focus on further optimizing the ICCP process, particularly in terms of catalyst regeneration strategies to mitigate deactivation.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.energyfuels.4c05631>.

DKR-350 material composition before and after sorting, DKR-350 separation efficiencies, complete mass balances of thermal and catalytic pyrolysis experiments, (semi)metal elemental analysis of pyrolysis oils, pilot-scale sorting plant overview, industrial shredder and loop dryer facility overview, pilot scale (catalytic) pyrolysis unit overview, 2D HSQC NMR spectra of dry-washed PO-rich stream pyrolysis oil and PO-poor stream catalytic pyrolysis oil, extended mono- and polyaromatic yields of the PO-poor catalytic pyrolysis, gas product yields of thermal and catalytic pyrolysis experiments, list of prevalent heterofunctional components in thermal and catalytic pyrolysis oils obtained from GC × GC-TOFMS, and detailed sorting protocol and schematic sorting protocol (PDF)

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Notes

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